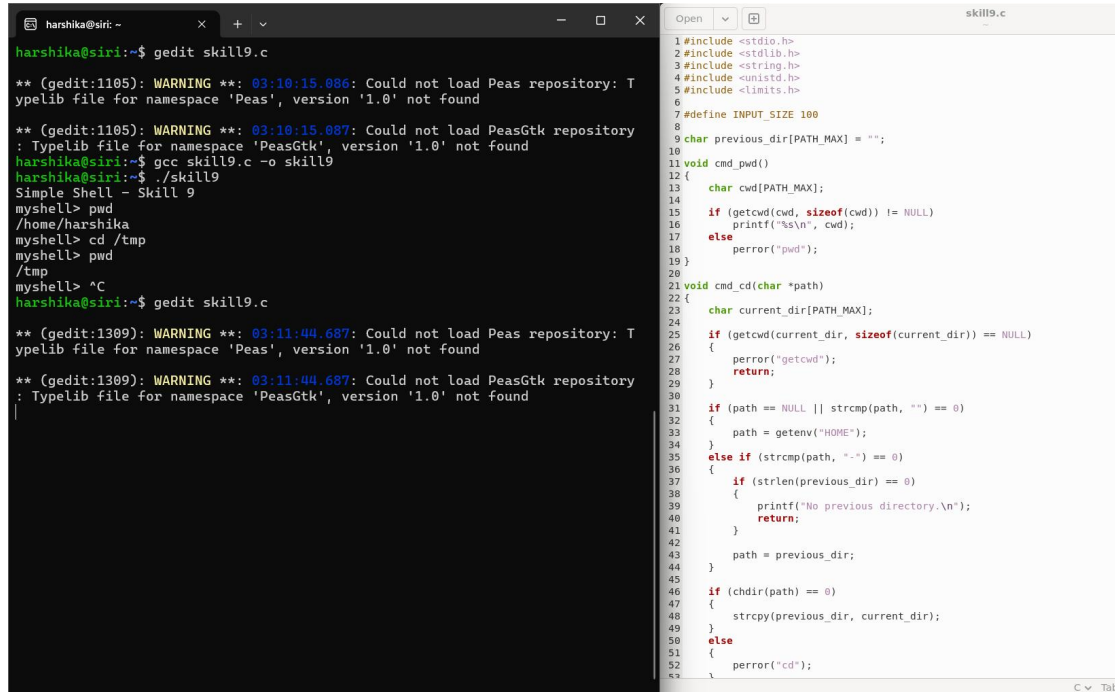


## Skill - 9

### 1. To change directories



The image shows a terminal window on the left and a code editor on the right, illustrating the implementation of a simple shell's directory change functionality.

**Terminal Window:**

```
harshika@siri: ~  
harshika@siri:~$ gedit skill9.c  
  
** (gedit:1105): WARNING **: 03:10:15.086: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found  
  
** (gedit:1105): WARNING **: 03:10:15.087: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found  
harshika@siri:~$ gcc skill9.c -o skill9  
harshika@siri:~$ ./skill9  
Simple Shell - Skill 9  
myshell> pwd  
/home/harshika  
myshell> cd /tmp  
myshell> pwd  
/tmp  
myshell> ^C  
harshika@siri:~$ gedit skill9.c  
  
** (gedit:1309): WARNING **: 03:11:44.687: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found  
  
** (gedit:1309): WARNING **: 03:11:44.687: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
```

**Code Editor (skill9.c):**

```
1 #include <stdio.h>  
2 #include <stdlib.h>  
3 #include <string.h>  
4 #include <unistd.h>  
5 #include <limits.h>  
6  
7 #define INPUT_SIZE 100  
8  
9 char previous_dir[PATH_MAX] = "";  
10  
11 void cmd_pwd()  
12 {  
13     char cwd[PATH_MAX];  
14  
15     if (getcwd(cwd, sizeof(cwd)) != NULL)  
16         printf("%s\n", cwd);  
17     else  
18         perror("pwd");  
19 }  
20  
21 void cmd_cd(char *path)  
22 {  
23     char current_dir[PATH_MAX];  
24  
25     if (getcwd(current_dir, sizeof(current_dir)) == NULL)  
26     {  
27         perror("getcwd");  
28         return;  
29     }  
30  
31     if (path == NULL || strcmp(path, "") == 0)  
32     {  
33         path = getenv("HOME");  
34     }  
35     else if (strcmp(path, "-") == 0)  
36     {  
37         if (strlen(previous_dir) == 0)  
38         {  
39             printf("No previous directory.\n");  
40             return;  
41         }  
42         path = previous_dir;  
43     }  
44  
45     if (chdir(path) == 0)  
46     {  
47         strcpy(previous_dir, current_dir);  
48     }  
49     else  
50     {  
51         perror("cd");  
52     }
```